PHYSICAL CHEMISTRY CUMULATIVE EXAMINATION

September 15, 1990

The photoelectron spectrum of PCl_3 is shown in the figure*. The spectrum was obtained using He(I) excitation, i.e., the exciting photon energy is 21.2 eV.

(A) Describe the principle of photoelectron spectroscopy.

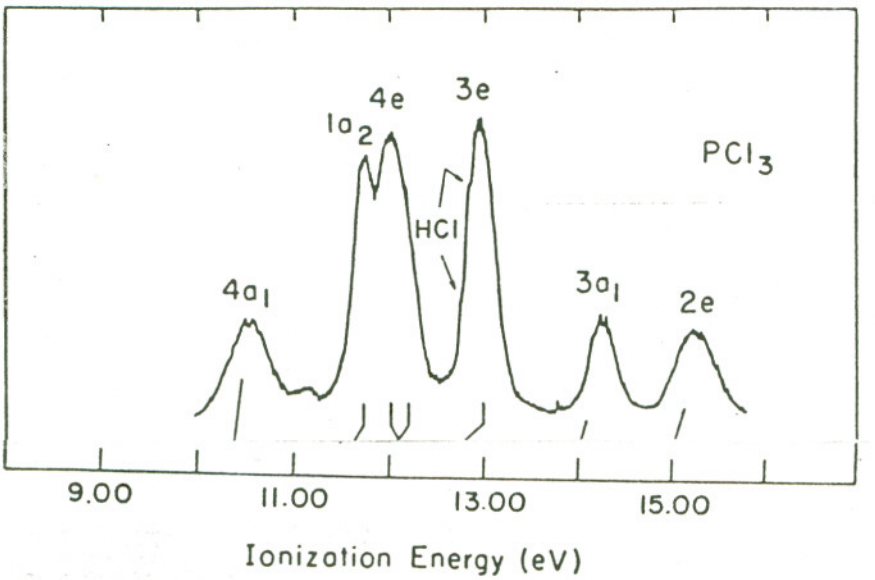
(B) Calculate the wave length of the exciting photons. Which spectral range would you assign these photons (e.g. visible, infrared, ultraviolet, extreme UV, x-ray, radio frequencies, etc.).

[1 eV = 1.6022 E-19 J; h = 6.62608 E-34 Js; c = 2.9979 E8 ms^{-1}]

(C) Calculate the velocity of the electron ejected from the highest (HOMO) molecular orbital upon ionization with a photon of 21.2 eV energy.

[Electron mass, m_e = 9.1094 E-31 kg]

(D) The peaks in the figure are assigned to molecular orbitals of various possible symmetry species under the molecular symmetry group, C_{3v}. Give the correct term designation for the radical ion left behind upon ionization, for each of the six ionizations shown in the figure.

(E) Some of the radical ion states created by ionizing PCl_3 cannot have C_{3v} equilibrium symmetry. Explain why.

* J.W. Rabalais, Principles of photoelectron spectroscopy, Wiley, 1977, p. 223.